DataVista Analytics & EcoWear Inc.

You have recently joined the **Data Visualization Division** at **DataVista Analytics**, a Colombo-based analytics and insights firm that provides outsourced data services to global clients. Your current project is with **EcoWear Inc.,** a U.S.-based outdoor sales company headquartered in Cascade Valley, a region well known for its dynamic weather patterns, ranging from snowy winters and heavy rains to long, sunny summers.

EcoWear caters to a multinational population, attracting both local residents and a large community of mobile professionals and international visitors. Previous analyses have shown that customer movement between regions often shifts with seasonal weather changes, influencing product demand in ways that are not always predictable.

EcoWear offers high-quality yet affordable outdoor lifestyle products, focusing on three core items:

- Product A – Winter Jacket
- Product B – Rain Boots
- Product C – Sunglasses

EcoWear has outsourced to DataVista Analytics to perform analysis and prediction on their data. They are currently in the initial phase, where they aim to identify patterns and trends in their existing datasets.

Part 1: Radar Chart (Weather vs. Product Performance)

Analyze how different weather factors influence the sales performance of three products and visualize these relationships using a radar chart.

Dataset Details:

- Generate your own dataset using the provided Python code **product_weather_dataset**. When creating it, provide the correct folder path in df_raw.to_csv() and df_norm.to_csv() to save the dataset; otherwise, it will be saved to the root folder by default.
- The dataset includes three products (A, B, C).
- Weather factors included in the dataset:
    - tavg – Average temperature (°C)
    - tmin – Minimum temperature (°C)
    - tmax – Maximum temperature (°C)
    - prcp – Precipitation (mm)
    - snow – Snowfall (cm)
    - wspd – Wind speed (km/h)
    - tsun – Sunny days (%)
    - pres – Atmospheric pressure (hPa)

Instructions for Your Analysis:

1. Dataset Preparation:

As a junior data analyst at DataVista, your first task is to study how weather factors influence product demand. Begin by generating your own weather-product impact dataset using the provided Python function, entering your student ID as student_index and any integer as random_seed to create a unique dataset.

You may run the code either locally or in Google Colab.

The function will produce two CSV files : a raw dataset and a normalized dataset, showing how factors like temperature, rainfall, snow, and sunshine affect the three products (winter jackets, rain boots, and sunglasses).

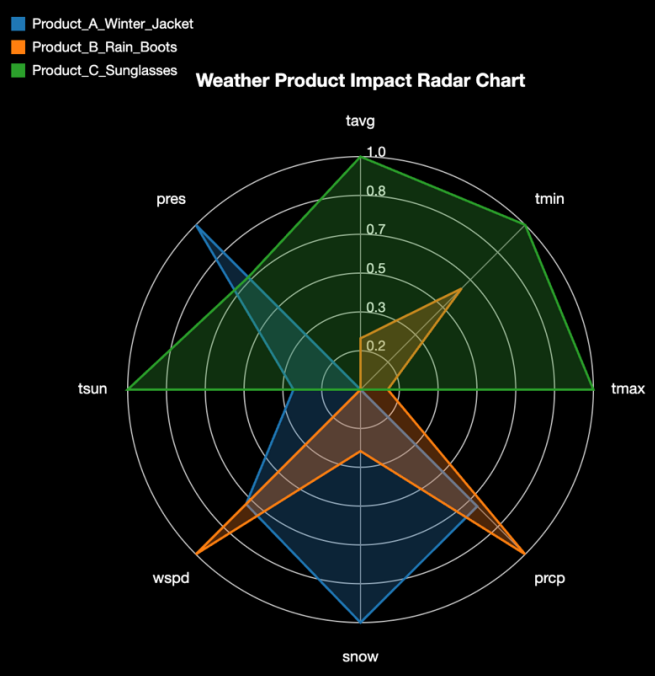
Use the most suitable dataset for the following analysis tasks.

2. Radar Chart Creation (Self-Study Task):

Use D3.js to create a radar chart comparing the three products across the selected weather factors using your chosen dataset. This task helps you build on the concepts learned during the practical session and further enhance your D3.js data visualization skills.

Follow these steps: *Set up SVG container → Define chart size and radius → Specify features (axes) → Load CSV data → Calculate angles for axes → Create color scale → Draw circular levels / background grid → Draw axes and axis labels → Create radar line generator (Key Concept) – Use d3.lineRadial() to convert data points into closed polygons; set .radius(d => d.value * radius), .angle((d,i) => i * angleSlice), and .curve(d3.curveLinearClosed) to automatically close the polygon → Draw radar polygons for each product – Map each product's feature values to points and draw the polygons with svg.append("path").datum(points).attr("d", radarLine); style with .attr("fill"), .attr("stroke"), and .attr("fill-opacity") → Add legend.*

Sample Output:



3. Chart Interpretation

After creating the radar chart, write a short interpretation (**maximum 250 words)** explaining the observed patterns, comparisons, or insights, such as which product performs best under certain weather conditions and how they differ across factors.

Include your interpretation directly below the chart **using D3.js's text()** function.

4. Deliverables

All deliverables related to this section must be submitted as outlined in the **Submission Guidelines**.

Part 2: Sankey Diagram (Customer Flow Analysis)

EcoWear wants to understand how customers move through their sales funnel, from initial website or app visits to purchase completion or drop-offs. The dataset tracks customer interactions at each stage, including acquisition channels (Organic Search, Paid Ads, Social Media), product page views, cart additions, checkout initiation, payments, order confirmations, and drop-offs. Each value in the dataset represents the number of customers in thousands moving from one stage to the next. The goal is to visualize these flows to identify stages with the highest drop-offs and stages with the highest conversions.

1. **Task Instructions**

- Use **D3.js** to create a **Sankey diagram** illustrating customer flows.
- Load the CSV dataset provided (EcoWear_Customer_Flow.csv) containing source, target, and value.
- Assign **different colors** for drop-offs versus successful conversions.
- Make links or nodes **interactive** (hover highlights, tooltip pop-ups).

2. **Deliverables**

- All deliverables related to this section must be submitted as outlined in the **Submission Guidelines.**

Part 3: Stacked Coxcomb Chart (Product Sales by Customer Segment)

EcoWear Inc. wants to understand **how each customer segment contributes to sales across all products**. The dataset contains sales figures (in thousands of units) for three key products(**Winter Jacket, Rain Boots, and Sunglasses**) broken down by **seven customer segments**: Local Residents, Mobile Professionals, International Visitors, Students, Senior Citizens, Tourists, and Corporate Clients.

The goal is to **visualize customer segments as angular slices**, with products stacked radially within each slice. This allows analysts to identify **which products are most popular for each segment** and compare contributions across segments.

1. **Task Instructions:**

- **Use D3.js to create a stacked coxcomb chart**. Each customer segment should form a slice (angular section) of the chart.
- **Dataset**: Use the CSV file EcoWear_Product_Sales.csv with columns: Product, CustomerSegment, Sales.
- **Stack products radially** within each slice according to their sales values.
- **Use distinct colors for each product** to maintain consistency.
- **Add interactivity**:
    - Tooltip on hover showing Customer Segment, Product, and Sales (in thousands).
- **Add a legend** identifying each product.

2. **Deliverables:**

- All deliverables related to this section must be submitted as outlined in the **Submission Guidelines.**

Part 4: Stacked Bar Chart (Monthly Sales by Product)

EcoWear Inc. wants to analyze overall monthly sales trends across their three core products: Winter Jacket, Rain Boots, and Sunglasses. The goal is to visualize how each product contributes to the total monthly sales, summing across all customer segments. This will help the company identify which products drive the most revenue each month and observe seasonal patterns.

Dataset Details:
The CSV file EcoWear_Monthly_Sales.csv contains four columns:

- Month : month of the year (January to December)
- Product : one of Winter Jacket, Rain Boots, or Sunglasses
- CustomerSegment : one of Local Residents, Mobile Professionals, International Visitors, Students, Senior Citizens, Tourists, or Corporate Clients
- Sales : monthly sales in thousands of units per product per customer segment

Note: *you need to sum the sales across all customer segments for each product in each month before plotting the stacked bars.*

1. Task Instructions:

- Use D3.js to create a stacked bar chart.
- X-axis: Months (January to December)
- Y-axis: Total sales (in thousands)

- Bars: Each bar represents a month and is stacked by product, with the total height representing total monthly sales.
- Tooltip: Show month, product, and total sales (summed across segments) on hover.
- Include a legend identifying each product.

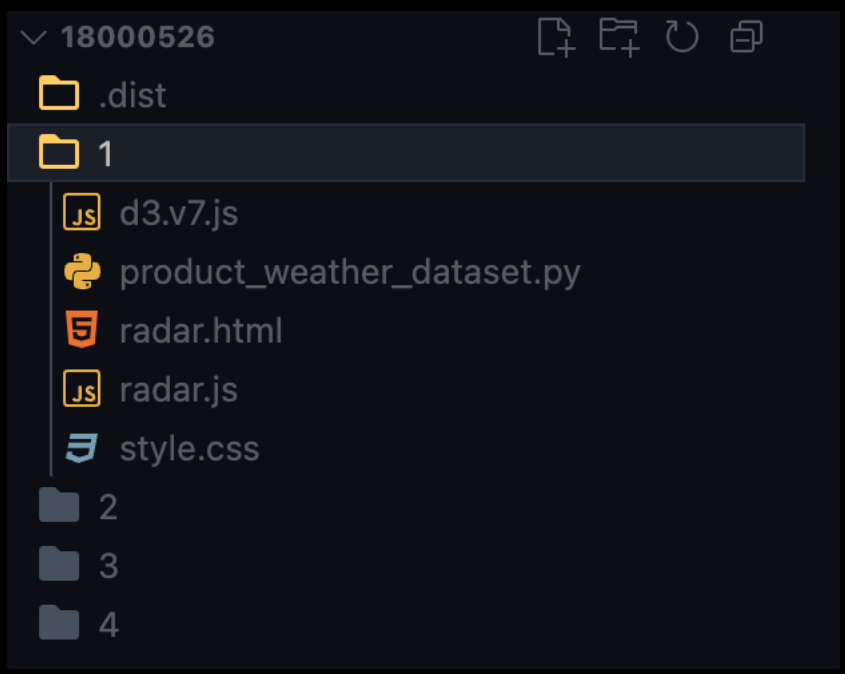
2. Deliverables:

- All deliverables related to this section must be submitted as outlined in the Submission Guidelines.

Part 5 : Submission Guidelines.

1. **Create a root folder** named with your **index number** (e.g., 18000526).
2. **Inside the root folder**, create **four subfolders**, one for each part:
    - 1 : Part 1: Radar Chart
    - 2 : Part 2: Sankey Diagram
    - 3 : Part 3: Stacked Coxcomb Chart
    - 4 : Part 4: Stacked Bar Chart
3. **Inside each subfolder**, include all relevant files for that part:
    - HTML file (e.g., index.html)
    - CSS file (if any styling is separate, e.g., style.css)
    - JS file (e.g., script.js) containing the D3.js code
    - CSV dataset file used for that part (e.g., EcoWear_Product_Sales.csv)

Example folder structure for index 18000526:



Important: *Ensure that all files are correctly named and located in the corresponding subfolders. This organization will help instructors evaluate each part efficiently.*

Dashboard Creation (All Charts Combined)

After completing Parts 1 to 4, you are required to create a dashboard that consolidates all four visualizations into a single, interactive view. This will help EcoWear Inc. get a comprehensive understanding of weather impacts, customer flows, customer segment contributions, and monthly sales trends.

1. Task Instructions
    a. Dashboard Layout:
        i. Use a single HTML page (e.g., dashboard.html) to host all charts.
        ii. Organize charts logically, so that the dashboard is clear and readable.
        iii. Ensure each chart retains its interactive features (tooltips, hover highlights, legends, etc.).
    b. Data Handling:
        i. Use the same CSV datasets provided in Parts 1–4.
        ii. You may load datasets using separate <script> files
    c. Interactivity & Usability:
        i. Maintain interactive features for each chart (tooltips, hover effects, legends).
        ii. Make the dashboard visually appealing with appropriate spacing, labels, and headings for each chart.
        iii.
2. Submission Files for Dashboard:

For the dashboard submission, you must include all required dashboard files inside the root folder named with your index number. Within this root folder, you must place **dashboard.html**, which displays all four charts, along with **dashboard.js** and **dashboard.css** (if you choose to use an external stylesheet).

The four individual parts (Part 1 to Part 4) must remain in their own subfolders (1, 2, 3, and 4), and each subfolder should contain the relevant HTML, CSS, JS, and CSV files for that part.

The **dashboard.html** file should load and render all four charts (Radar Chart, Sankey Diagram, Stacked Coxcomb Chart, and Stacked Bar Chart) by using **dashboard.js**, which must include the implementations for all four visualizations.

You must ensure that **dashboard.html**, **dashboard.js**, and **dashboard.css** are placed directly inside the root folder (index number) as part of the final submission.